

Measuring the Potential of Artificial Intelligence to Assist Local Governance

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1. Introduction

Local governance is responsible for public services that influence the everyday quality of life, from education to sanitation. Its effectiveness depends on officials to implement regulations, and on citizens to understand them. However, like any institution, local governments suffer from numerous issues. These include outdated zoning laws that have historically reinforced segregation and social inequality¹, difficulty in finding information without extensive research, and complicated regulations overlapping across jurisdictions from town to county to state levels². These illustrate the challenges of improving the structures and processes of local governance, and highlight the information burden for citizens, officials, and policymakers. Zoning in particular is a relevant case because its regulations span multiple jurisdictions, use technical language, and directly impact local communities.

This study considers the role of Artificial Intelligence (AI), specifically Large Language Models (LLMs). LLMs are AI systems trained on enormous text datasets to predict and generate language³. Owing to their ability to process large volumes of technical text, they are used in various sectors, including healthcare and finance, to assist with tasks and improve efficiency⁴.

¹ Flint, Anthony. "Reforming a Century-Old Approach to Land Use." *Lincoln Institute of Land Policy*, 22 Dec. 2022, www.lincolnst.edu/publications/articles/2022-12-state-local-zoning-reform/.

² Lens, Michael C. "Zoning, Land Use, and the Reproduction of Urban Inequality." *Annual Review of Sociology*, U.S. National Library of Medicine, 2022, [pmc.ncbi.nlm.nih.gov/articles/PMC10691857/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC10691857/).

³ Zhao, Wayne Xin, et al. "A Survey of Large Language Models." *arXiv.Org*, 31 Mar. 2023, arxiv.org/abs/2303.18223.

⁴ Raza, Mubashar, et al. "Industrial Applications of Large Language Models." *Scientific Reports*, U.S. National Library of Medicine, 21 Apr. 2025, [pmc.ncbi.nlm.nih.gov/articles/PMC12012124/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC12012124/).

LLMs are not immune to errors, one of which is the tendency to hallucinate and fabricate false information⁵. This risk must be considered for platforms that use them.

Given the capabilities and risks of LLMs, this thesis examines the extent to which LLMs can interpret local zoning laws and answer questions, and how access to legal and administrative documents affects their performance. The hypothesis is that providing a locally run LLM with domain-specific legal and administrative documents will improve answer quality and reduce hallucinations compared to responses generated without access to those documents.

Using the zoning laws and administrative bodies applicable in Madison, WI, as a case study, the thesis tests the performance of a low-resource, locally run LLM with and without access to source documents. The study found measurable improvement in the answer quality once LLMs were given access to domain-relevant documents; however, this did not guarantee the elimination of hallucinations. Based on these findings, this thesis argues that LLMs have the potential to enhance local governments as assistive tools supported by sufficient safeguards.

2. Background and Conceptual Framework

2.1 Artificial Intelligence and Large Language Models

Artificial intelligence (AI) can be broadly defined as computer systems designed to perform tasks that usually require human intelligence⁶. One notable type is the Large Language Model (LLM), a type of neural network-based AI capable of performing diverse linguistic tasks by analyzing text data. LLMs identify relationships between token sequences and compute probabilities, enabling tasks such as language translation, summarization, and content generation⁷. LLMs are relevant to local governance as related tasks involve the interpretation and application of textual materials in the form of regulations and administrative procedures.

2.2 Local Governance in the Context of Planning and Zoning

⁵ “Large Language Models Pose Risk to Science with False Answers, Says Oxford Study.” *University of Oxford*, 20 Nov. 2023, www.ox.ac.uk/news/2023-11-20-large-language-models-pose-risk-science-false-answers-says-oxford-study

⁶ Gignac, Gilles, et al. “Defining Intelligence: Bridging the gap between human and artificial perspectives” *Intelligence*, vol. 104, 2024, <https://www.sciencedirect.com/science/article/pii/S0160289624000266>.

⁷ Idan, Daphna, and Sharon Einav. “Primer on Large Language Models: An Educational Overview for Intensivists.” *Critical Care*, vol. 29, p. 238, 2025, pmc.ncbi.nlm.nih.gov/articles/PMC12164094/.

Local governance in this thesis refers to the system of rules, institutions, and decision-making processes through which local communities are planned, regulated, and administered at the county, city, town, and village levels. For this study, local governance is examined through the local bodies involved in zoning, along with the city and county ordinances, and state statutes that define their powers. Zoning refers to the use of the public regulatory power, or police power, to specify how land may be used and developed⁸. Zoning is a relevant case to study as it operates across multiple jurisdictions and layers of authority that directly shape housing access and daily life for citizens, developers, and local officials⁹. It is a useful case for LLMs as it allows a combination of both theory and application-based questions to be tested.

2.3 Planning and Zoning Institutions in Wisconsin

The local bodies examined are the Plan Commission and Zoning Board, which play important roles in zoning and planning at the local level. Under Wisconsin law, the Plan Commission is charged with making and adopting a master plan for the development of the municipality, and its work supports the governing body's land-use and development decisions¹⁰. The Plan Commission reviews ordinance consistency and makes recommendations on planning. The Zoning Board's role is more of a quasi-judicial body, which reviews and decides cases where a relaxation of the ordinance is sought or where there is an alleged error in a zoning decision. Zoning Boards do not change local zoning laws. It functions like a court and must follow state laws and local zoning ordinances¹¹. For zoning regulations, there are legal regulations across multiple jurisdictions that a developer, Zoning Board, and Plan Commission must adhere to. This study refers to the ordinances of the city of Madison, WI¹², the ordinances of Dane County¹³, and the statutes set forth by the State Government of Wisconsin¹⁴. The categories and domains vary

⁸ Markham, Lynn, and Rebecca Roberts. Zoning Board Handbook: For Wisconsin Zoning Boards of Adjustment and Appeals. 2nd ed., University of Wisconsin-Madison Extension, 2006, fyi.extension.wisc.edu/landusetraining/files/2020/01/Zoning_Board_Handbook-condensed.pdf.

⁹ Jenny Schuetz. "Is Zoning a Useful Tool or a Regulatory Barrier?" The Brookings Institution, 31 Oct. 2019, www.brookings.edu/articles/is-zoning-a-useful-tool-or-a-regulatory-barrier/.

¹⁰ Wisconsin Legislature. Wisconsin Statutes § 62.23: City Planning. Wisconsin Legislative Documents, docs.legis.wisconsin.gov/statutes/statutes/62/i/23.

¹¹ University of Wisconsin-Madison, Center for Land Use Education. "Zoning Board Basics." Land Use Training & Resources, UW-Madison Extension, fyi.extension.wisc.edu/landusetraining/zoning-board-basics/.

¹² City of Madison. Code of Ordinances. Madison, WI, Municode Library, library.municode.com/wi/madison/codes/code_of_ordinances

¹³ Dane County. Ordinances. Dane County, Wisconsin, www.danecounty.gov/ordinances

¹⁴ Wisconsin Legislature. Wisconsin Statutes and Annotations. Wisconsin Legislative Documents, docs.legis.wisconsin.gov/statutes/statutes

in structure and complexity, allowing the opportunity to test if there is a difference in LLM performance across explanatory administrative text and technical legal text.

3. Research Design

3.1 Research Question and Hypothesis

This study examines the extent to which Large Language Models (LLMs) can interpret local zoning laws and answer questions, and how access to legal and administrative documents affects their performance. It evaluates whether providing relevant materials improves answer quality and reduces the amount of hallucination in responses on local government administration and regulations. It connects these findings to practical implementations for local governments.

The central research question is: To what extent does providing a locally run LLM with domain-specific legal and administrative documents improve answer quality and reduce hallucinations when responding to local governance and zoning-related questions?

The hypothesis is that providing a locally run LLM with domain-specific legal and administrative documents will improve answer quality and reduce hallucinations compared to responses generated without access to those documents.

The treatment variable in this study was access to domain-specific documents. In the without-data condition, the LLM received only the question and general prompt instructions. In the with-data condition, the LLM received the same question and prompt instructions, and access to relevant documents for that domain. For the mixed category-specific question sets, the LLM was provided with documents of both categories.

The unit of analysis was the individual question-response pair. Each question was answered once in the without-data condition and once in the with-data condition in a within-subjects design, making the question serve as its own comparison case¹⁵. The dependent variables are the answer-quality score and hallucination score.

3.2 Study Objective

The setting aimed to replicate an environment where a user with minimal technical expertise and financial resources could set up a local system to prompt LLMs with specific questions on

¹⁵ Newsom, Jason T. “Within-Subjects/Repeated-Measures/Paired t Test.” Univariate Statistics, Portland State University, web.pdx.edu/~newsomj/uvclass/ho_within.pdf.

zoning and zoning laws applicable in Madison, WI, and provide data for the LLM to reference. Rather than testing the performance of a more advanced subscription-based or specialized LLM, the study evaluated a freely accessible open-weight model running on consumer hardware, making the study closer to a test of practical feasibility than an upper-bound estimate of AI performance.

3.3 Technical Environment

For this study, the LLM used was Ollama's llama3.1:8b¹⁶ model released by Meta AI. Instead of providing the data and question files in a chat with an LLM through a website, the setup also tested the technical constraints of running an LLM on an individual system by downloading it and running it locally. The laptop used was an HP Victus 15 with 16 GB RAM. The LLM was accessed through a Python script that submitted an Excel file with questions listed and recorded the responses in the same spreadsheet. Each domain used the same general prompt structure. The prompt templates are provided in Appendix B.

This is an open-weight model, meaning the model weights are accessible¹⁷, in contrast to models like ChatGPT, which is accessed through the provider. The LLM was instruction-tuned, which means that the model is trained to answer like an assistant and perform tasks for the user¹⁸, replicating a practical environment for LLM application. The knowledge cutoff was December 2023, which meant the LLM was unaware of any events or legislation after that date¹⁹. While there is no definitive scale to measure LLM intelligence, there are various benchmarks, one of which is the Massive Multitask Language Understanding (MMLU) benchmark, a collection of over 15,000 questions on various subjects, including elementary mathematics, US history, computer science, law, and more²⁰. The LLM used in the study has an MMLU score of 69.4%, meaning it correctly answered 69.4% of the questions. For comparison, ChatGPT's GPT-4o, released in 2024, scored 88.7%²¹. These scores are only meant to provide a general context for

¹⁶ Ollama. "llama3.1:8b." Ollama Model Library, Ollama, ollama.com/library/llama3.1:8b

¹⁷ Zhang, Gongbo et al. "Closing the gap between open source and commercial large language models for medical evidence summarization." *NPJ digital medicine* vol. 7,1 239. 9 Sep. 2024, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11383939/>

¹⁸ Zhang, Shengyu, et al. "Instruction Tuning for Large Language Models: A Survey." arXiv, 14 Mar. 2024, arxiv.org/abs/2308.10792v5

¹⁹ NVIDIA. "Meta Llama 3.1 8B – NeMo." NVIDIA NGC Catalog, NVIDIA, catalog.ngc.nvidia.com/orgs/nvidia/teams/nemo/models/llama-3_1-8b-nemo?version=1.0

²⁰ Hendrycks, Dan, et al. "Measuring Massive Multitask Language Understanding." arXiv, 12 Jan. 2021, arxiv.org/abs/2009.03300

²¹ OpenAI. "Hello GPT-4o." OpenAI, 13 May 2024, www.openai.com/index/hello-gpt-4o/.

LLM intelligence. There was no additional fine-tuning or testing done prior to the study. The study did not change any of Ollama’s default values.

3.4 Data Sources

The questions are divided into two main categories, consisting of different domains. One category is for administrative duties and functions relating to the Plan Commission and Zoning Board domains. The other category is laws and regulations within the domains of the City of Madison, Dane County, and the State of Wisconsin. The question sets tested each domain separately with and without access to source documents. Each category was also tested separately with a mix of questions from the different domains, with the dataset of both categories available for the LLM to access. These questions were only tested with the treatment variable and are reported as supplemental evidence. The dataset for the administration category consisted of the Plan Commission and Zoning Board Handbooks. The dataset for the law category consisted of select Madison City Ordinances, Dane County Ordinances, and Wisconsin State Statutes.

Domain	Category	Source Material	Data Format
State Statutes	Law	Wisconsin State Statutes	PDF
County Ordinances	Law	Dane County Ordinances	PDF
City Ordinances	Law	Madison City Ordinances	Word Document
Plan Commission	Administration	Plan Commission Handbook	PDF
Zoning Board	Administration	Zoning Handbook	PDF

Table 1. Summary of data sources, domain, category, and format.

3.5 Question Construction

All questions were constructed from the information in the dataset, ensuring that the LLM would not require additional references to answer the questions. The questions were factual, procedural, and scenario-based to accurately represent the different types of queries that a chatbot on zoning would get. Factual questions included identifying and explaining information directly stated in the administrative handbooks and laws. Procedural questions tested the responsibilities and process of different boards and agencies involved in zoning. Scenario-based questions tested the

proper execution of rules and regulations given a situation. The number of questions varied by domain because the source materials differed in length and complexity.

Question	Type of Question	Domain
What does Dane County define as a Shoreland?	Factual	County Ordinances
If a zoning board member wants to change the ordinance, what should they do?	Procedural	Zoning Board
Can I get a zoning certificate to build a 15,000 foot data center in Madison?	Scenario	City Ordinances

Table 2. Examples of types of questions

3.6 Evaluation Rubric

The LLM responses were assessed using two rubric-based scoring systems: answer quality and hallucination score. The “Answer Quality” corresponds to the correctness and completeness of the answer provided. This was measured on a five-point scale: 5 = very good, 4 = good, 3 = fair, 2 = poor, and 1 = very poor.

The “Hallucination Score” measures the amount of hallucinated content in the answer. Hallucination refers to the generation of text that is factually incorrect, logically inconsistent, or entirely fabricated²². An answer is considered to have had some degree of hallucination if it cites laws and regulations that do not exist. Hallucination was also scored on a five-level rubric, with higher scores corresponding to greater accuracy and fewer hallucinations: 1 = no hallucinations or misinformation, 0.75 = very little outdated or incorrect information, 0.5 = some outdated or incorrect information, 0.25 = mostly outdated or incorrect information, and 0 = hallucination or false information. A higher hallucination score corresponds to a lower number of hallucinations in the answer. The full scoring rubric and examples are provided in Appendix A.

Each response was manually evaluated using the answer-quality and hallucination rubrics. Answer quality and hallucination were scored separately, as an answer can be incomplete without fabricating information. A single evaluator conducted the scoring, and the results should be interpreted with the limitation that inter-rater reliability was not assessed. Statistical analysis

²² Anh-Hoang, Dang et al. “Survey and analysis of hallucinations in large language models: attribution to prompting strategies or model behavior.” *Frontiers in artificial intelligence* vol. 8 1622292. 30 Sep. 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12518350/>

and visualization of scores were conducted in R. The calculated statistical outputs are provided in Appendix C.

4. Methods of Analysis

4.1 Paired Pre/Post Design

Since the questions were evaluated in two conditions, first the “without-data” condition and then the “with-data” condition, this study used a paired pre/post design. In paired designs, the same unit is measured twice, so the analysis focuses on the difference between conditions²³. The unit of analysis was the individual question-response pair. The design reduces variation across questions as each serves as its own case for comparison.

4.2 Condition Averages

In a paired design, these averages show the overall level of performance in each condition, while the difference between them provides an initial indication of whether the treatment variable was associated with improvement.

4.3 Wilcoxon Signed-Rank Test

By treating the “with-data” condition as a binary experimental condition, the Wilcoxon signed-rank test was used to compute p-values for paired comparisons of answer quality and hallucination. The Wilcoxon signed-rank test is a nonparametric test for paired data, used to assess if the median of the paired differences differs from zero²⁴. The test was run separately for each domain and outcome variable. By convention, a p-value less than 0.05 is taken as statistically significant.

4.4 Standardized Mean Paired Difference

To measure how the improvement in performance compares to normal variation across questions, the study calculates the standardized mean paired difference, which quantifies the size

²³ Parab, Shraddha, and Supriya Bhalerao. “Choosing statistical test.” International journal of Ayurveda research vol. 1,3 (2010): 187-91. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2996580/>

²⁴ Rosner, Bernard et al. “The Wilcoxon signed rank test for paired comparisons of clustered data.” Biometrics vol. 62,1 (2006) <https://pubmed.ncbi.nlm.nih.gov/16542245/>

of an effect relative to variability, allowing researchers to distinguish between effects that are merely statistically detectable and effects that are substantively large²⁵. This was calculated separately by domain as the mean paired gain divided by the standard deviation of the paired gains for each outcome. Positive values indicate that the with-data condition outperformed the without-data condition. Larger positive values indicate a stronger effect.

4.5 Rate of Improvement

The improvement rate was calculated as the percentage of questions for which the with-data score was higher than the without-data score. Ties were not counted as improvements. Improvement rates were calculated separately for answer quality and hallucination scores. This measured whether, even for questions the LLM was able to answer well, introducing data still improved the answer quality or lowered the amount of hallucination.

4.6 95% Bootstrap Confidence Intervals

Finally, the study reported 95% bootstrap percentile confidence intervals for the mean paired gain in each domain. The bootstrap is a resampling method used to estimate uncertainty around a statistic, including means and confidence intervals, by repeatedly resampling from the observed data²⁶. These intervals are interpreted as showing a plausible range for the true average gain. If the entire interval lies above zero, that supports a positive treatment effect. Narrow intervals indicate greater precision, while wider intervals indicate more uncertainty²⁷. The analysis used 2000 bootstrap resamples for each domain.

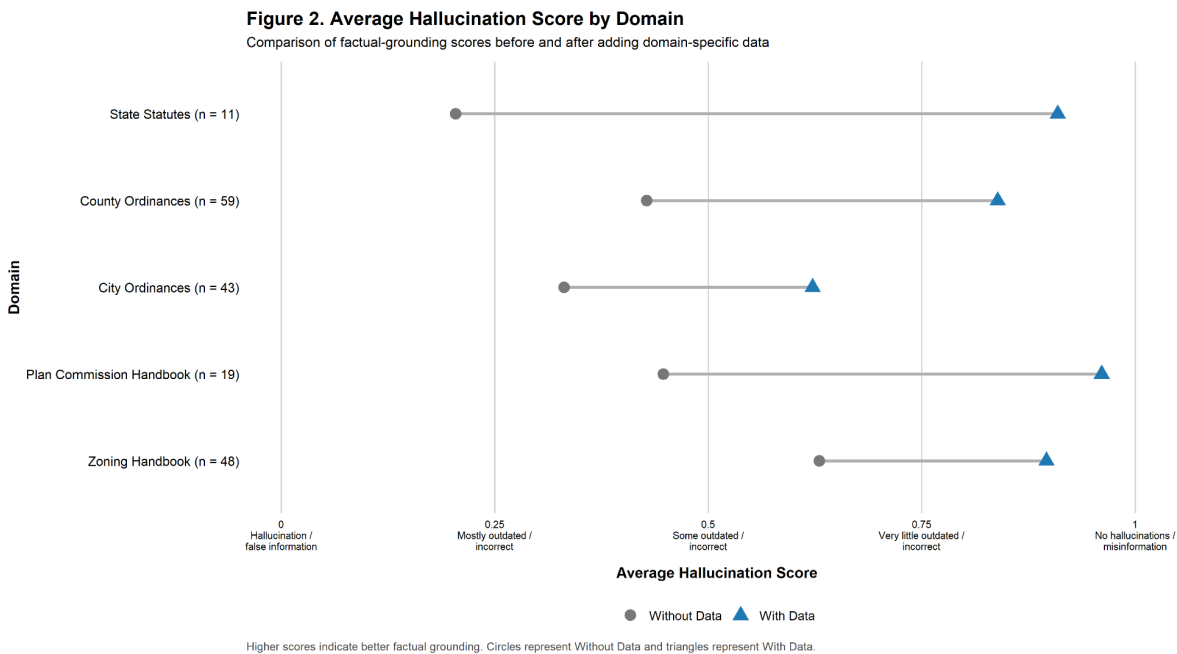
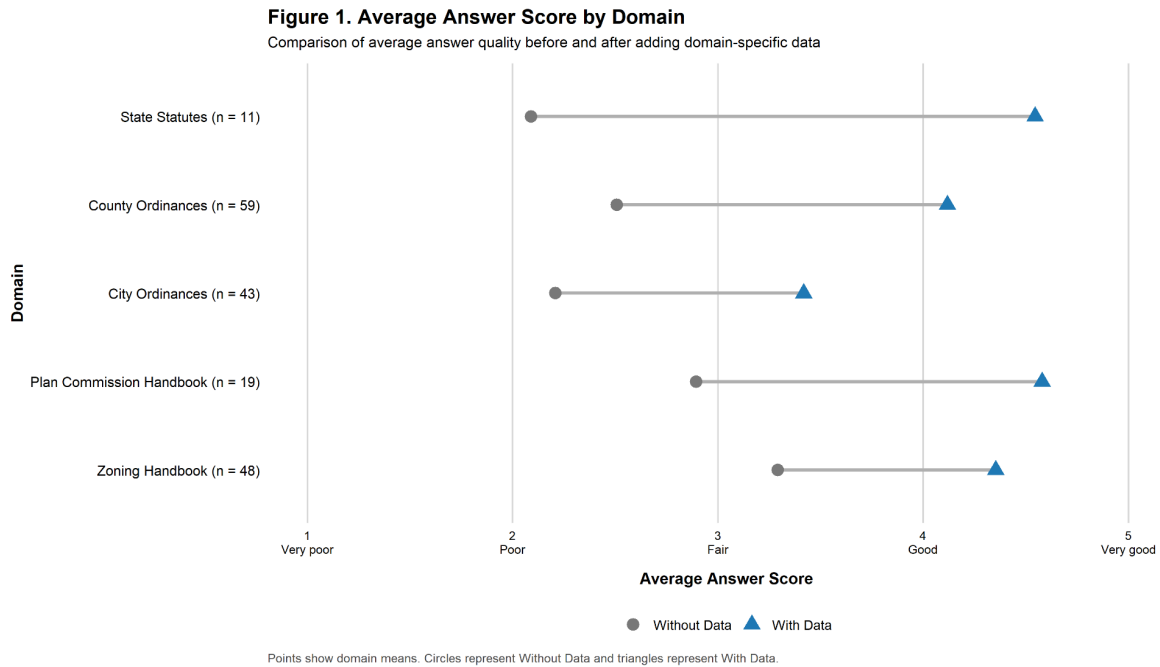
5. Results

5.1 Changes in Average Scores Across Domains

²⁵ Smith, Lindsey J. Wolff, and S. Natasha Beretvas. “Estimation of the Standardized Mean Difference for Repeated Measures Designs.” *Journal of Modern Applied Statistical Methods*, vol. 8, no. 2, 1 Nov. 2009 <https://digitalcommons.wayne.edu/cgi/viewcontent.cgi?article=1537&context=jmasm>

²⁶ Rousselet, Guillaume A., Cyril R. Pernet, and Rand R. Wilcox. “The Percentile Bootstrap: A Primer With Step-by-Step Instructions in R.” *Advances in Methods and Practices in Psychological Science*, vol. 4, no. 1, Jan.–Mar. 2021, pp. 1–10, <https://www.pure.ed.ac.uk/ws/portalfiles/portal/203604809/2515245920911881.pdf>

²⁷ National Institute of Standards and Technology (NIST). “1.3.5. Quantitative Techniques: Confidence Intervals for the Mean.” NIST/SEMATECH e-Handbook of Statistical Methods, NIST, www.itl.nist.gov/div898/handbook/eda/section3/eda352.htm.



Across all domains, the average answer quality and hallucination score improved. With data, the average score was an answer rating of 4.09 or “Good”, a clear improvement from an average of 2.66, or between “Poor” and “Fair”, in the without-data condition, a gain of 1.43 on the 5-point rubric. The with-data condition improved the average hallucination score in all domains, reaching 0.819, between “No Hallucinations or Misinformation” and “Very Little

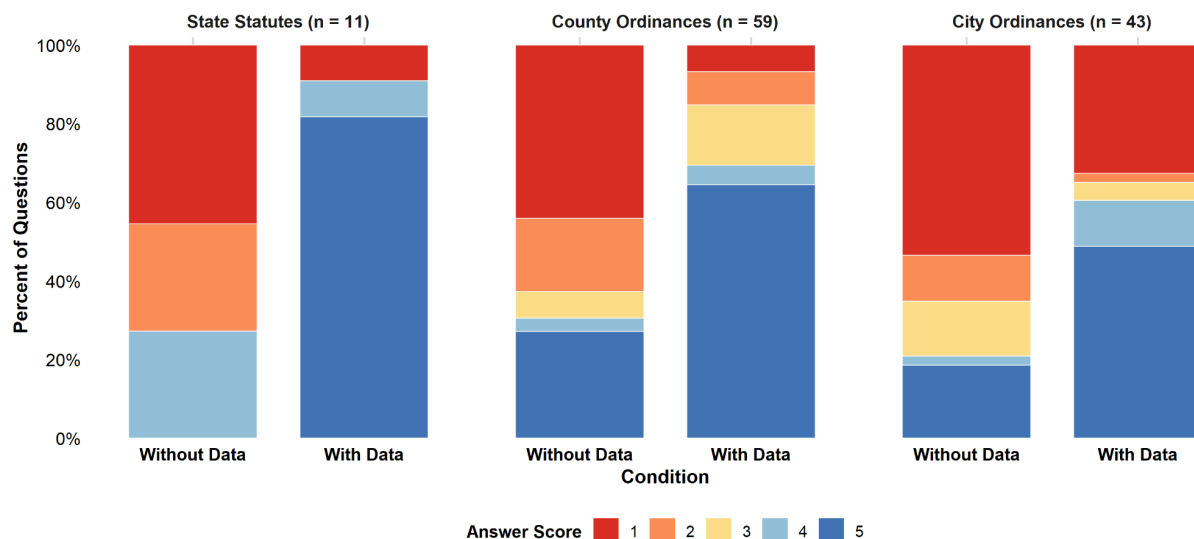
Outdated/Wrong Information” from a score of 0.447, between “Some Outdated/Wrong Information“ and “Mostly Outdated/Wrong Information” without data. Most domains improved to hallucination scores above 0.89, with the exception of County Ordinances (0.839) and City Ordinances (0.622).

There were also category-specific mixed question sets for law and administration, which were a mix of questions from domains of that category. These sets were tested only in the with-data condition to evaluate whether the LLM could answer questions if all documents were available at once. The law set scored an average of 4.54, and the administrative set’s average was 4.42, both averages between “very good” and “good” on the rubric. The hallucination scores were also high, with the administrative answers having an average of 0.92, and the law category having an average of 0.875.

5.2 Distributional Changes in Scores

Figure 3. Distribution of Answer Scores: Law and Ordinances

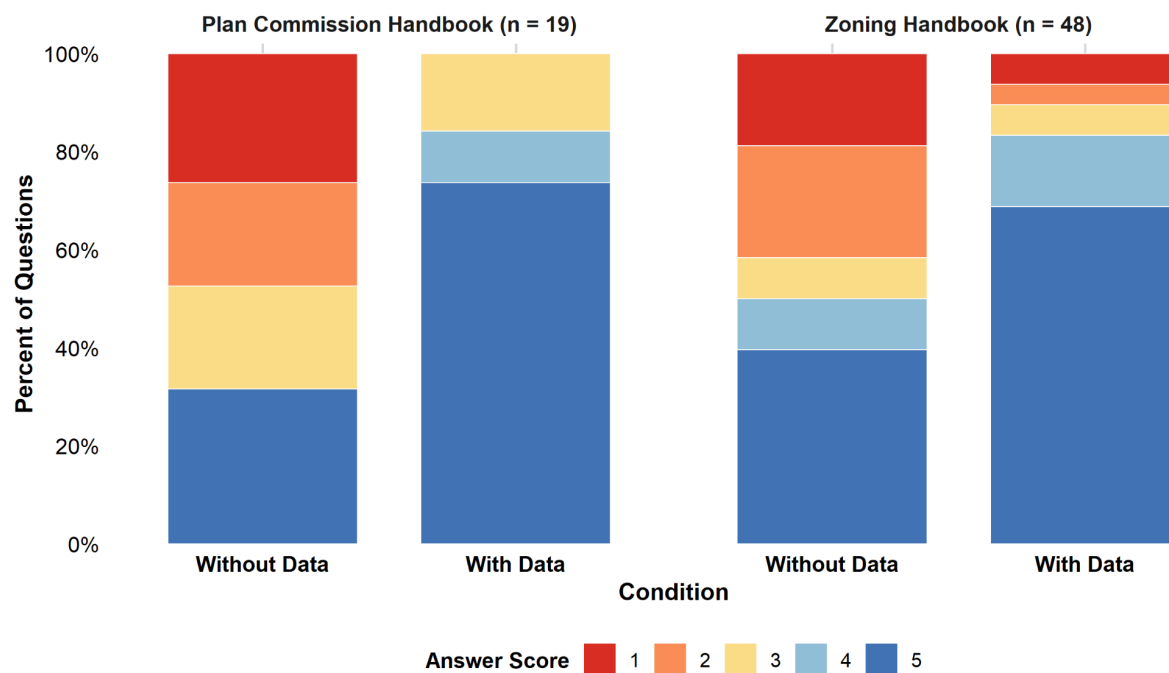
State Statutes, County Ordinances, and City Ordinances



Each bar shows the percentage distribution of answer-quality scores within a domain and condition.

Figure 4. Distribution of Answer Scores: Administrative Bodies

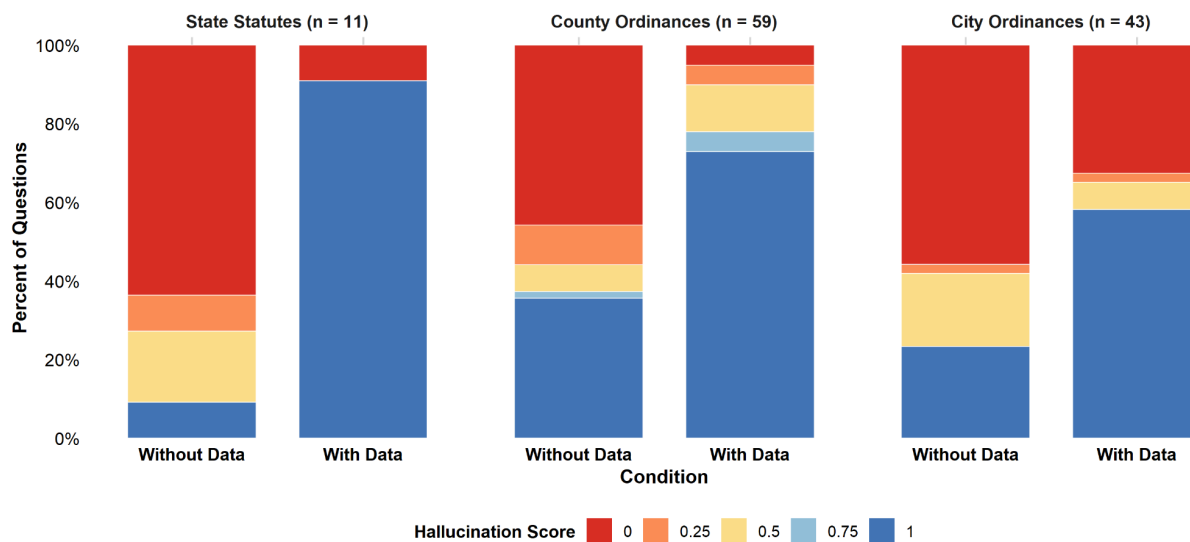
Plan Commission Handbook and Zoning Handbook



Each bar shows the percentage distribution of answer-quality scores within a domain and condition.

Figure 5. Distribution of Hallucination Scores: Law and Ordinances

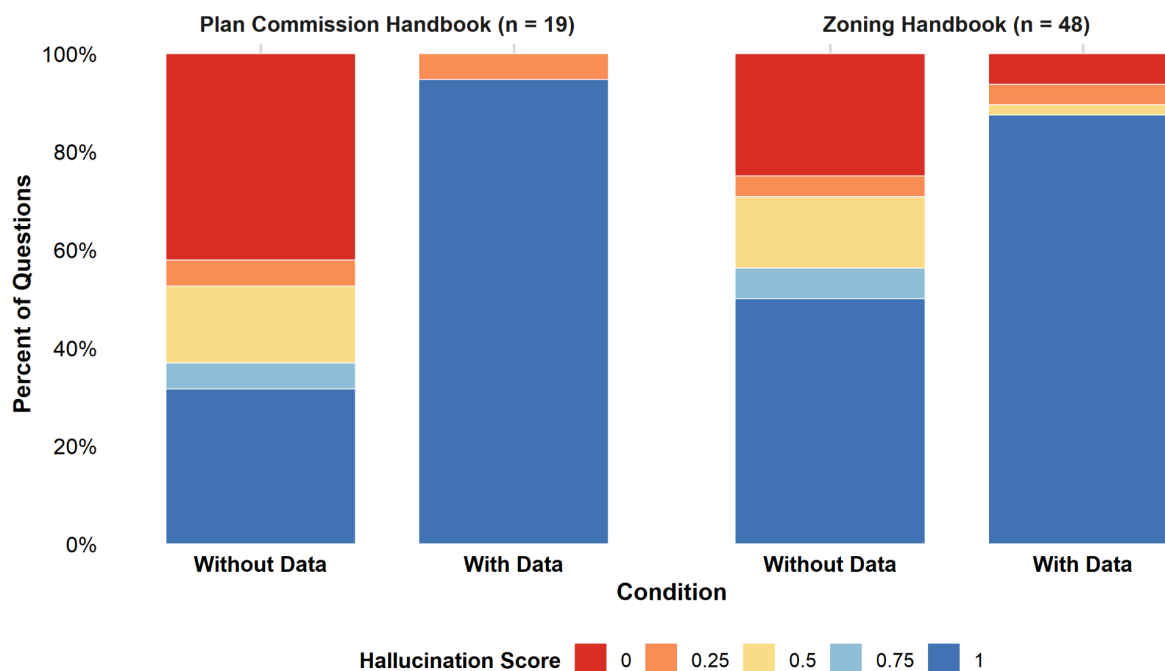
State Statutes, County Ordinances, and City Ordinances



Higher hallucination scores indicate stronger factual grounding and less misinformation.

Figure 6. Distribution of Hallucination Scores: Administrative Bodies

Plan Commission Handbook and Zoning Handbook



Higher hallucination scores indicate stronger factual grounding and less misinformation.

The distribution shows that in the with-data condition, more responses have higher answer quality scores and hallucination scores. This is especially visible in State Statutes and the administrative handbooks. However, the results also suggest that data access did not eliminate serious errors such as hallucinations. This suggests that the LLM still had difficulty with some questions despite documents provided, like in the case of City Ordinances, where over a quarter of the answers are graded the lowest quality in the with-data condition.

5.3 Statistical Significance of the Treatment Effect

The Wilcoxon signed-rank tests showed statistically significant gains across all domains for both answer quality and hallucination scores. The p-values for answer quality and hallucination across all domains were under 0.05, suggesting that the treatment variable was associated with a statistically significant improvement.

5.4 Magnitude of Improvement Across Domains

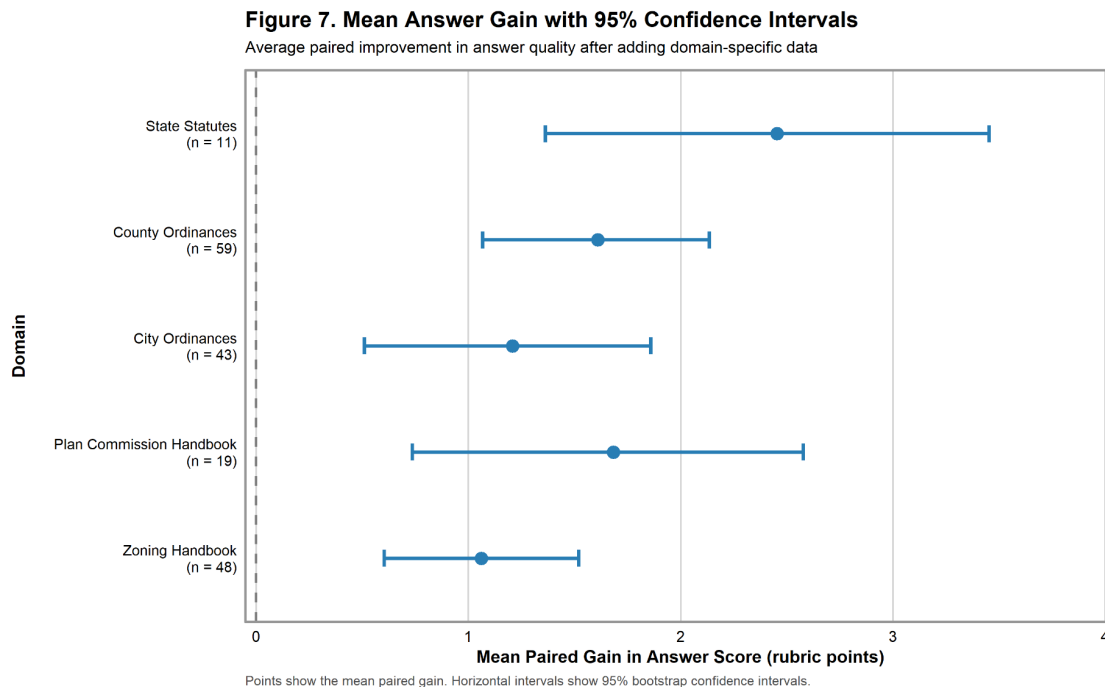
The standardized effect sizes show that the magnitude of improvement varied by domain. State Statutes had the largest standardized gains, while City Ordinances had the smallest. The domains followed an identical rank order by standardized effect size in both the answer-quality and

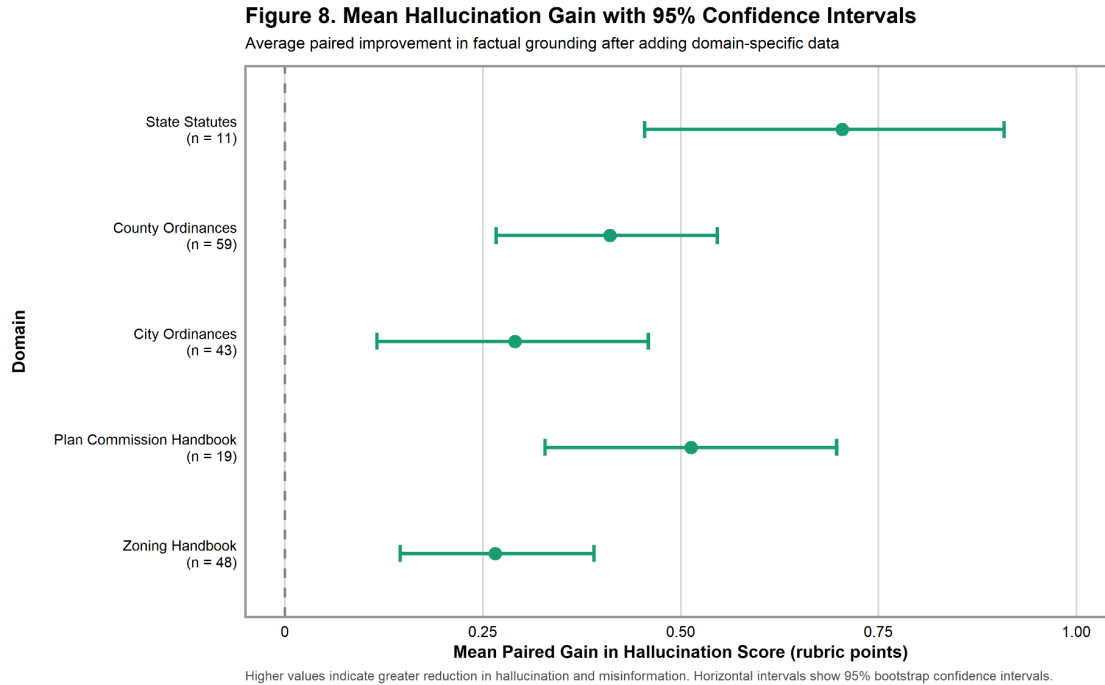
hallucination scores, showing improvement was not limited to one measure of performance. The order from most to least was: State Statutes, Plan Commission, County Ordinances, Zoning Board, City Ordinances.

5.5 Practical Improvement Rates

The size of improvement varied by domain, but in all domains, at least half of the questions received improved answers with data. The domain with the highest percentage of answer improvements was in State Statutes, with 81.8% of questions having a higher score across the with-data condition than without, while the lowest was 51.2% for City Ordinances. The hallucination score improvement also varied by domain, with State Statutes improving the most for 81.8% scores, and the Zoning Board with the lowest improvement percentage at 43.8%. However, it is important to note that this domain was the best performing in answer quality and hallucination scores, explaining why the rate may be lower, as there is less to improve.

5.6 Confidence Intervals and Uncertainty





Among all domains, State Statutes had the greatest improvement in answer quality. The estimated mean gain was 2.45 rubric points, with a 95% confidence interval of 1.36 to 3.45, indicating a consistent positive improvement in answer quality. This suggests that the with-data condition improved responses by approximately one-and-a-half to three-and-a-half rubric categories, with the point estimate being about two-and-a-half rubric categories. The other domains also displayed positive gains of at least one rubric point on average. Although City Ordinances also showed improvement, their lower confidence bound of about 0.5 rubric points indicates a smaller effect.

Like answer quality, State Statutes had the greatest improvement in hallucination scores. The estimated mean gain was 0.7 rubric points, with a 95% confidence interval of 0.45 to 0.9, indicating a consistent positive improvement in how often the LLM hallucinates, with a score of 1 indicating no hallucinations at all. The other domains also displayed positive gains of at least 0.25 points on the one-point rubric.

6. Interpretation of Findings

6.1 Overall Interpretation

The results suggest that providing domain-specific documents was associated with statistically significant improvements in a locally run LLM, improving answer accuracy and reducing hallucination in responses. In every domain, the same questions in the with-data condition led to higher answer quality scores. This pattern was consistent across distributional shifts, statistical significance tests, standardized effect sizes, improvement rates, and confidence intervals. The treatment variable was associated with a statistically significant improvement for all categories. The results by domain followed an identical rank order by standardized effect size in both the answer-quality and hallucination scores, suggesting that the relative strength of the treatment effect was consistent across both measures.

The improvements did not guarantee reliability, as the inclusion of data did not stop the occurrence of hallucinations. A concerning pattern was that the law category had a high number of hallucinations, highlighting potential risks if LLMs were to be treated as a source for laws, as there is a chance the answer is hallucinated. No category or domain was able to achieve perfect hallucination scores on all answers, even with access to data. As a result, while data enhances performance, it does not eliminate the need for verification, and LLMs should be understood as assistive tools rather than definitive sources of information.

6.2 Why Performance Varied Across Domains

The dataset files were used by the LLM in their original downloaded format and not converted to more desirable formats, such as Markdown²⁸, to best replicate an environment in which someone without technical expertise may implement it. One possible explanation for the performance of City Ordinances may be that it was the only data in Word Document format, while the rest were in PDF, which may have led to less accurate information retrieval from the provided data. Another explanation is that some of the City Ordinance questions required referrals to tables included in the documents, which is why the LLM struggled with information retrieval and cross-referencing²⁹.

6.3 Limits of the Study

The setting aimed to replicate an environment where someone with minimal technical expertise and financial resources could set up a local system to prompt LLMs. The study was only performed using one free-to-download LLM to treat its performance closer to a general baseline compared to premium models with more resources and higher costs. The exact version used was

²⁸ Chen, Zhongpu, et al. “MDEval: Evaluating and Enhancing Markdown Awareness in Large Language Models.” arXiv, 27 Aug. 2025, arxiv.org/abs/2501.15000

²⁹ Tian, Shi-Yu, et al. “TabularMath: Understanding Math Reasoning over Tables with Large Language Models.” arXiv, 26 May 2025, arxiv.org/abs/2505.19563.

llama3.1:8b, an 8 billion parameter model, the lowest parameter model of Llama 3.1, as the higher parameter models would have struggled to run on a personal laptop. The knowledge cutoff was December 2023, which meant the LLM was unaware of any events or legislation after that date. There was no additional fine-tuning or testing done prior to the study. In an official deployment by an organization or government, it is likely the LLM would be fine-tuned before deployment, which would affect the quality of answers and frequency of hallucination. Fine-tuned LLMs are associated with higher levels of performance³⁰ and allow the model to be personalized and prepared based on the user's needs³¹. These limitations suggest that the results should be interpreted as a baseline for low-resource LLM deployment rather than an upper bound on performance.

The study was also limited to materials from Madison, Dane County, and Wisconsin. The results may differ in other jurisdictions, policy areas, or legal domains. Future work should include reviews from professionals to validate scores. Since the questions were constructed from the source materials, the study only evaluates performance in document-grounded questions and not the full range of questions that might be asked in practice. A further limitation is that all responses were manually scored by a single evaluator, so inter-rater reliability was not assessed. Improvement in performance may come through greater computational resources, improved data processing, and model fine-tuning.

7. Practical Strategies and Applications for Local Governments

7.1 What Local Governments Must Consider When Implementing AI

7.1.1 Technical Expertise

AI may provide rapid answers to questions, but to effectively utilize the abilities of AI, local governments require a degree of technical expertise³². As the study showed, the baseline knowledge and abilities of freely accessible AI systems have limits, which means that local governments will have to invest to enhance the abilities of any AI they implement. This would

³⁰ Tian, Katherine, et al. "Fine-Tuning Language Models for Factualty." arXiv, 14 Nov. 2023, arxiv.org/abs/2311.08401

³¹ Clarke, Christopher, et al. "PEFT-U: Parameter-Efficient Fine-Tuning for User Personalization." arXiv, 25 July 2024, arxiv.org/abs/2407.18078.

³² App Maisters Inc. "Generative AI in Government: Opportunities and Risks.", App Maisters, gov.appmaisters.com/generative-ai-in-government-opportunities-and-risks/.

include fine-tuning the model based on the requirements³³ and proper data processing³⁴. Local governments can outsource this work. With sufficient resources and time, the goal of using AI to enhance and assist local governance is realistic.

7.1.2 Accuracy

There must be standards for ensuring the accuracy of any AI deployed³⁵. This would involve having a process that tests the accuracy of the proposed product before and after deployment to ensure the quality does not decline over time and regular updates to prepare AI for any changes in policy to avoid generating false and outdated outputs.

7.1.3 Risk Management

If AI is used by local governments, there must be sufficient risk assessments and structures in place to ensure the capabilities necessary to address any problems that may arise from the AI. An example of a framework a local government may adopt is the National Institute of Standards and Technology's AI Risk Management Framework. The framework provides a structured approach for managing risks associated with artificial intelligence systems throughout their lifecycle. It organizes AI risk management around four core functions: Govern, Map, Measure, and Manage. Govern includes establishing review structures and training employees in risk fluency. Mapping requires creating inventories of AI systems, tracking their use, including the data they utilize, the decisions they influence, and the regulations that apply. Measure is to assess how risk manifests, observing if models exhibit bias and how stakeholders can be affected. Manage is to establish risk mechanisms like access controls for sensitive data and monitoring for degradation in quality and accuracy³⁶. By following safe implementation strategies, local governments can mitigate risk and still harness the potential of artificial intelligence to safely support their work.

7.2 Citizen Assistance

³³ Lu, Wei, et al. "Fine-Tuning Large Language Models for Domain Adaptation: Exploration of Training Strategies, Scaling, Model Merging and Synergistic Capabilities." arXiv, 5 Sept. 2024, arxiv.org/abs/2409.03444.

³⁴ Budach, Lukas, et al. "The Effects of Data Quality on Machine Learning Performance." arXiv, 29 July 2022, arxiv.org/abs/2207.14529.

³⁵ National Institute of Standards and Technology (NIST). "New Report Details Challenges of Monitoring Deployed AI Systems." NIST Newsroom, 9 Mar. 2026, www.nist.gov/news-events/news/2026/03/new-report-challenges-monitoring-deployed-ai-systems.

³⁶ National Institute of Standards and Technology (NIST). "AI Risk Management Framework (AI RMF 1.0)." Information Technology Laboratory – Artificial Intelligence Risk Management Framework, NIST, www.nist.gov/itl/ai-risk-management-framework.

LLMs can help citizens by answering general questions about regulations and services. For example, in the case of zoning, if someone wanted to build a structure near Dane County Airport, an LLM could inform the user if there are any limitations on structure dimensions. One example of a city using LLMs for a similar purpose is Covington, Kentucky, which launched a chatbot called “Clive”, which answers questions about zoning and economic development³⁷. It is powered by GPT-4o, for a cost of less than \$200, and was provided with 4.6 billion characters of data from more than 17 PDF files, other documents, and 174 web links. “Clive” stands out as an example of Artificial Intelligence assisting local governments. These chatbots for FAQs would be required to remain up to date with changes in laws and policies, but have the potential to reduce administrative burden, specifically learning costs by using chatbots to inform and address queries accurately³⁸.

As shown by the case of Covington, this is a practical application as local governments can outsource the technical work, removing any potential for issues arising from hardware constraints and lack of expertise. To ensure accuracy, the LLMs have to be updated when policy changes to prevent them from crafting their responses based on outdated data, leading to inaccurate responses. The accuracy of the LLM has to be verified periodically to ensure the LLM is not providing outdated information. Hallucination is another important consideration. In a setting where an LLM would be trying to help a citizen solve issues and find the right resources, this is a major risk as the burden to cross-check falls on the user who is using the LLM for assistance in the first place. Organizations use a disclaimer to warn the user that AI may hallucinate, but this does not address the problem of misinformation by hallucination.

7.3 Policymaking and Loophole Detection

Although this study did not directly test policy drafting or loophole detection, the findings suggest that LLMs may have value as tools for policy research. LLMs can potentially assist with policymaking by finding loopholes in existing policies and providing recommendations for new policies. The aim would be to give policymakers ideas for improvement. This could be explored through hybrid AI, which is an experimental system that combines multiple methods like data-driven models and rule-based reasoning into a single working design in order to test and

³⁷ Davidson, Nikki. “LLM on a Budget: How Small Cities Are Leveraging AI.” Government Technology, 12 Nov. 2024, www.govtech.com/biz/data/llm-on-a-budget-how-small-cities-are-leveraging-ai.

³⁸ Lavoie-Gagne, Ophelie et al. “Artificial Intelligence as a Tool to Mitigate Administrative Burden, Optimize Billing, Reduce Insurance- and Credentialing-Related Expenses, and Improve Quality Assurance Within Health Care Systems.” *Arthroscopy : the journal of arthroscopic & related surgery : official publication of the Arthroscopy Association of North America and the International Arthroscopy Association* vol. 41,8 (2025). <https://pubmed.ncbi.nlm.nih.gov/40120727/>

refine approaches to a problem³⁹. Recent research argues that with sufficient computational resources, such systems could help publicly evaluate new legislation and reveal tax gaps stemming from loopholes⁴⁰. These systems would require technical expertise to assist in training and maintaining these systems. This application could be used to help improve laws and address loopholes, creating more efficient policies. If AI were to be used by a policymaker to try to come up with policies, there is still the ability to cross-check and verify if the LLM is hallucinating. This is an experimental application that can be improved upon, making it an application that would remain relevant if utilized successfully. Because this study tested legal knowledge and application rather than policy drafting, this should be treated as a possible extension.

7.4 Other Potential Applications

Another application beyond the scope of the study is the use of algorithms. At the local level, governments have begun to explore incorporating algorithms and LLMs to solve localized issues. For example, the city of Pittsburgh plans to use LLMs to analyze Pittsburgh's streets, identify which intersections are the highest-risk for accidents, and work with city officials to test new street design improvements⁴¹. The incorporation of AI through algorithms has the potential to address issues, but it is not risk-free. An example is a child care benefits scandal in the Netherlands, where a self-learning AI system wrongfully accused families of childcare welfare fraud⁴². This example illustrates why AI systems require oversight and careful monitoring for bias or error.

8. Conclusion

This study found that low-resource, locally run LLMs demonstrated measurable capacity to interpret local zoning laws and answer questions. Access to legal and administrative documents improved the answer quality and reduced hallucination and misinformation. However, the improvements did not guarantee reliability, as hallucinations persisted.

³⁹ Zhang, Hongbo, et al. "A Hybrid Prototype Method Combining Physical Models and Generative Artificial Intelligence to Support Creativity in Conceptual Design." *ACM Transactions on Computer-Human Interaction*, vol. 31, no. 5, article 65, Nov. 2024. <https://dl.acm.org/doi/full/10.1145/3689433>

⁴⁰ Fratrič, Peter, Nils Holzenberger, and Diego R. Amariles. "Can AI Expose Tax Loopholes? Towards a New Generation of Legal Policy Assistants." *arXiv*, 21 Mar. 2025, arxiv.org/abs/2503.17339.

⁴¹ Carnegie Mellon University College of Engineering. "Safe AI for Safer Streets for All." *Engineering News*, 10 Oct. 2024, engineering.cmu.edu/news-events/news/2024/10/10-safer-streets.html.

⁴² Bayer, Lili. "Dutch Scandal Serves as a Warning for Europe over Risks of Using Algorithms." *Politico*, 29 Mar. 2022, www.politico.eu/article/dutch-scandal-serves-as-a-warning-for-europe-over-risks-of-using-algorithms/.

This study argues that local governments, with sufficient technical expertise and safeguards, can use forms of AI such as LLMs as assistive tools. LLMs can make complex information more accessible and potentially assist policymakers, but they should not be implemented without oversight.

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Appendix

Appendix A: Scoring Rubrics

A.1 Answer Quality

Score	Label	Description
5	Very good	The answer is correct and complete.
4	Good	The answer is correct but missing some detail.
3	Fair	The answer is generally correct but missing an important element.
2	Poor	Misleading answer with few correct elements.
1	Very poor	Mostly incorrect and irrelevant.

A.2 Hallucination Score

Score	Label		Description
1	No hallucinations or misinformation		The answer is factual.
0.75	Very little outdated or incorrect information		The answer is mostly factual with one incorrect detail.
0.5	Some outdated or incorrect information		The answer is partially factual and partially incorrect.
0.25	Mostly outdated or incorrect information		The answer is mostly wrong but has one or two correct details.
0	Hallucination or false information		The answer is completely hallucinated and references nonexistent information.

Appendix B: Prompt Templates

B.1 Single-Domain Prompt Template:

"Your assistance is required with answering some questions related to [DOMAIN/TOPIC]"

"You will be given an excel of questions under the header 'Question' and you will fill answers in 'Model Answer'. "

"You are an assistant for zoning data and laws. "

"When you answer each question, ONLY GIVE THE ANSWER—be direct and to the point."

B.2 Single-Domain Prompt Variations

Domain	Domain-Specific Prompt
Zoning Board	"Your assistance is required with answering some questions related to Zoning in Madison, Wisconsin. "
Plan Commission	"Your assistance is required with answering some questions related to the Plan Commission in Dane County. "
City Ordinances	"Your assistance is required with answering some questions related to City Ordinances in the city of Madison, Wisconsin. "
County Ordinances	"Your assistance is required with answering some questions related to Zoning in Dane County. "
State Statutes	"Your assistance is required with answering some questions related to State Statutes in Wisconsin. "

B.3 Mixed-Category Prompt Template:

"Your assistance is required with answering some questions related to [TOPIC]"

"You will be given an excel of questions under the header 'Question' and you will fill answers in 'Model Answer'. "

"You are an assistant for zoning data and laws. "

"When you answer each question, ONLY GIVE THE ANSWER—be direct and to the point."

B.4 Mixed-Category Prompt Variations:

Category	Prompt
Administrative (Zoning Board + Plan Commission)	"Your assistance is required with answering some questions related to the Plan Commission and Zoning Handbook for the city of Madison, Wisconsin. "
Laws (City Ordinances + County Ordinances + State Statutes)	"Your assistance is required with answering some questions related to the state statutes, county and city ordinances for the city of Madison, Wisconsin. "

Appendix C: Statistical Outputs and Calculated Values

C.1 Domain Averages by Condition

Domain	Without-Data Answer Mean	With-Data Answer Mean	Without-Data Hallucination Mean	With-Data Hallucination Mean
Zoning Board	3.291666667	4.354166667	0.630208333	0.895833333
Plan Commission	2.894736842	4.578947368	0.447368421	0.960526316
City Ordinances	2.209302326	3.418604651	0.331395349	0.622093023
County Ordinances	2.508474576	4.118644068	0.427966102	0.838983051
State Statutes	2.090909091	4.545454545	0.204545455	0.909090909

C.2 Mixed-Category Averages

Category	Answer Mean	Hallucination Mean
Law	4.541666667	0.875
Administrative	4.421052632	0.921052632

C.3 Mean Paired Gains

Domain	Mean Answer Gain	Mean Hallucination Gain
Zoning Board	1.0625	0.265625
Plan Commission	1.684210526	0.513157895
City Ordinances	1.209302326	0.290697674
County Ordinances	1.610169492	0.411016949
State Statutes	2.454545455	0.704545455

C.4 Wilcoxon Signed-Rank Test Results

Domain	Answer Quality p-value	Hallucination p-value
Zoning Board	0.000185525	0.000388664
Plan Commission	0.004434704	0.001350153
City Ordinances	0.005382253	0.005619888
County Ordinances	0.00000489997	0.00000417511
State Statutes	0.009579798	0.007086405

C.5 Standardized Effect Sizes

Domain	Answer Quality Effect Size	Hallucination Effect Size
Zoning Board	0.632060124	0.622751781
Plan Commission	0.808523188	1.174826705
City Ordinances	0.53938476	0.499219688
County Ordinances	0.772424627	0.757067965
State Statutes	1.317151528	1.760113999

C.6 Rate of Improvement

Domain	Answer Improvement Rate	Hallucination Improvement Rate
Zoning Board	0.520833333	0.4375
Plan Commission	0.631578947	0.684210526
City Ordinances	0.511627907	0.488372093
County Ordinances	0.627118644	0.576271186
State Statutes	0.818181818	0.818181818

C.7 Bootstrap Confidence Intervals

Domain	Metric	Mean Gain	95% CI Lower	95% CI Upper
Zoning Board	Answer	1.0625	0.604166667	1.520833333
Plan Commission	Answer	1.684210526	0.736842105	2.578947368
City Ordinances	Answer	1.209302326	0.512214033	1.860465116
County Ordinances	Answer	1.610169492	1.06779661	2.13559322
State Statutes	Answer	2.454545455	1.363636364	3.454545455
Zoning Board	Hallucination	0.265625	0.145833333	0.390625
Plan Commission	Hallucination	0.513157895	0.328947368	0.697368421
City Ordinances	Hallucination	0.290697674	0.11627907	0.459302326
County Ordinances	Hallucination	0.411016949	0.266949153	0.546610169
State Statutes	Hallucination	0.704545455	0.454545455	0.909090909

Appendix D: Research Materials

Research Folder:

https://drive.google.com/drive/folders/10VaGsW2hCNuZBfeJew3kgQePupPXiux?usp=drive_link

D1. Folder Structure

Folder	Content
00_README	Brief guide to the folder.
01_Prompts	Prompts used in the study.
02_Question_Files	Original question files submitted to the LLM.
03_Answered_Files	LLM responses and manual scores.
04_Master_Dataset	Combined master scoring file.
05_R_Analysis	R script and outputs.